

Verification of Gamma-Ray Emission from the Crab Nebula and Cassiopeia A Using the Gammapy Analysis Package



Nicholas Bo You Chin*, David Kieda* for the VERITAS Collaboration†
Department of Physics & Astronomy, University of Utah, Salt Lake City, Utah*
veritas.sao.arizona.edu †

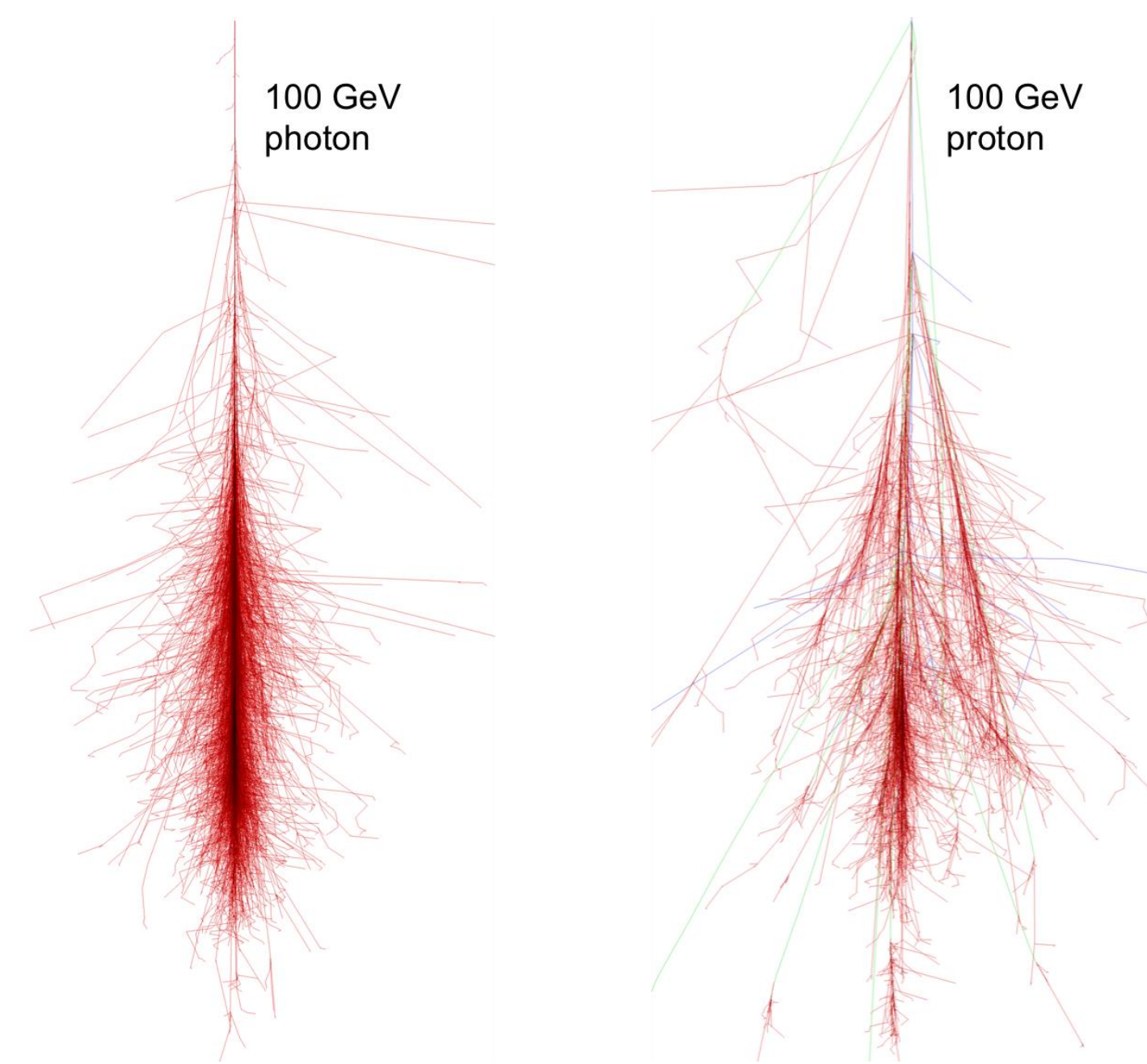
Gamma-Ray Astronomy observes and studies the highest-energy form of electromagnetic radiation emitted by celestial objects and phenomena. It allows us to investigate extreme environments and energetic events, including supernova remnants, pulsars, blazars, black holes, neutron stars, galaxy clusters, and diffuse gamma-ray emission throughout the Milky Way.¹



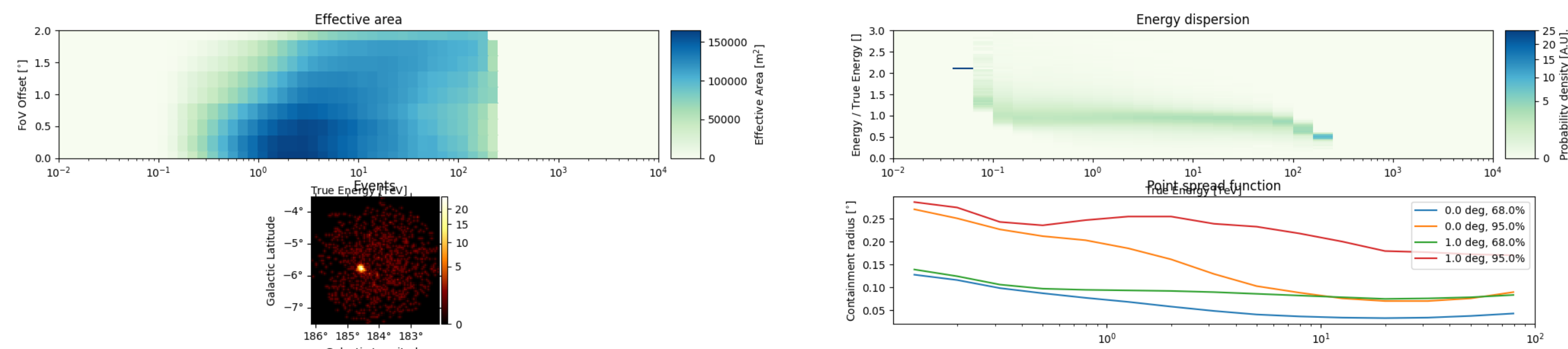
Image of the four telescopes at VERITAS

VERITAS

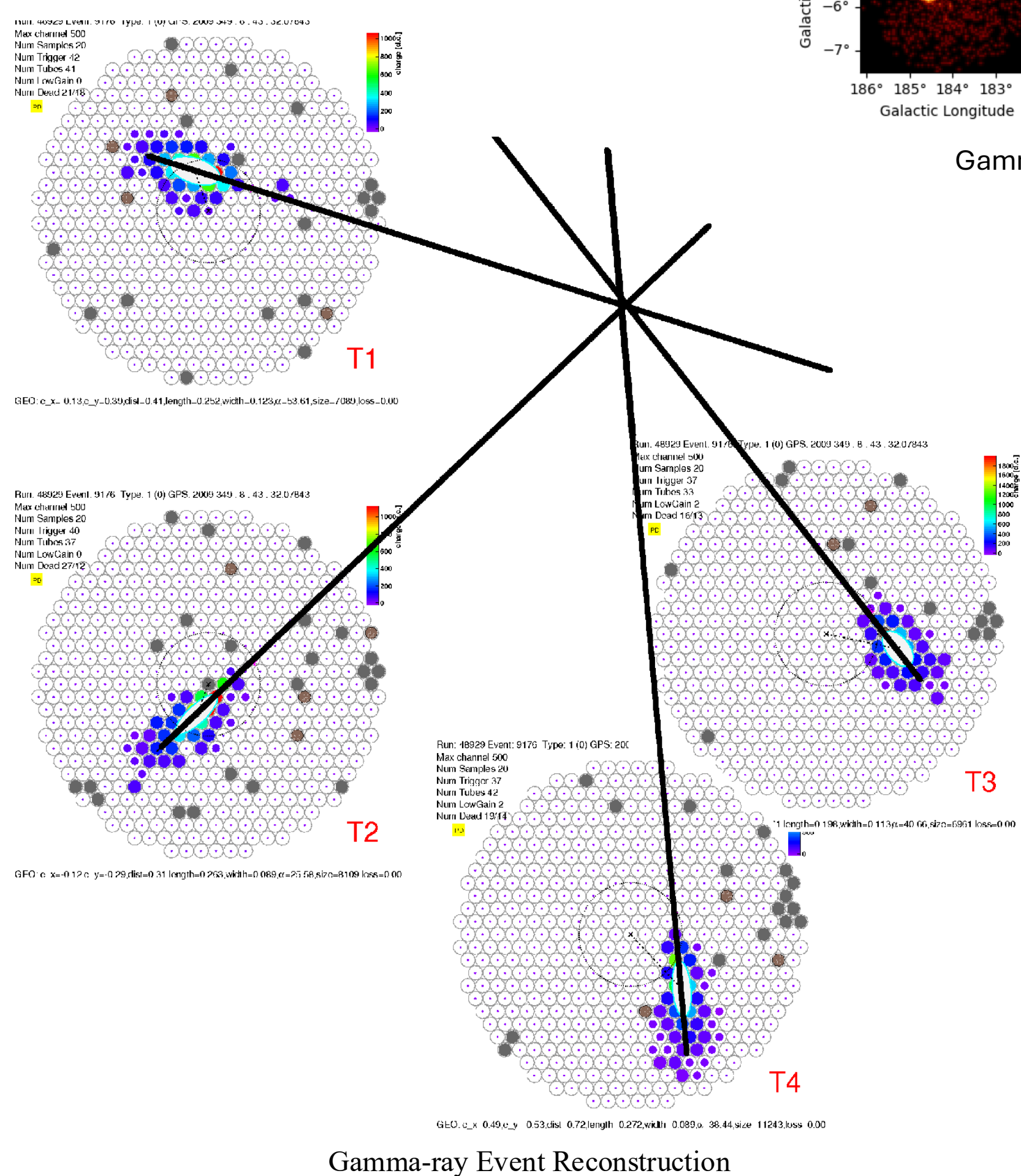
(Very Energetic Radiation Imaging Telescope Array System) is a ground-based gamma-ray observatory with four 12 meter wide optical reflectors for GeV-TeV photon energy range gamma-ray astronomy.³ It uses Imaging Atmospheric Cherenkov Telescope technique which allows determination of the gamma-ray induced extensive air shower core position and the precise reconstruction of the shower geometry.



Monte Carlo Simulations of Photon and Protons induced showers.²



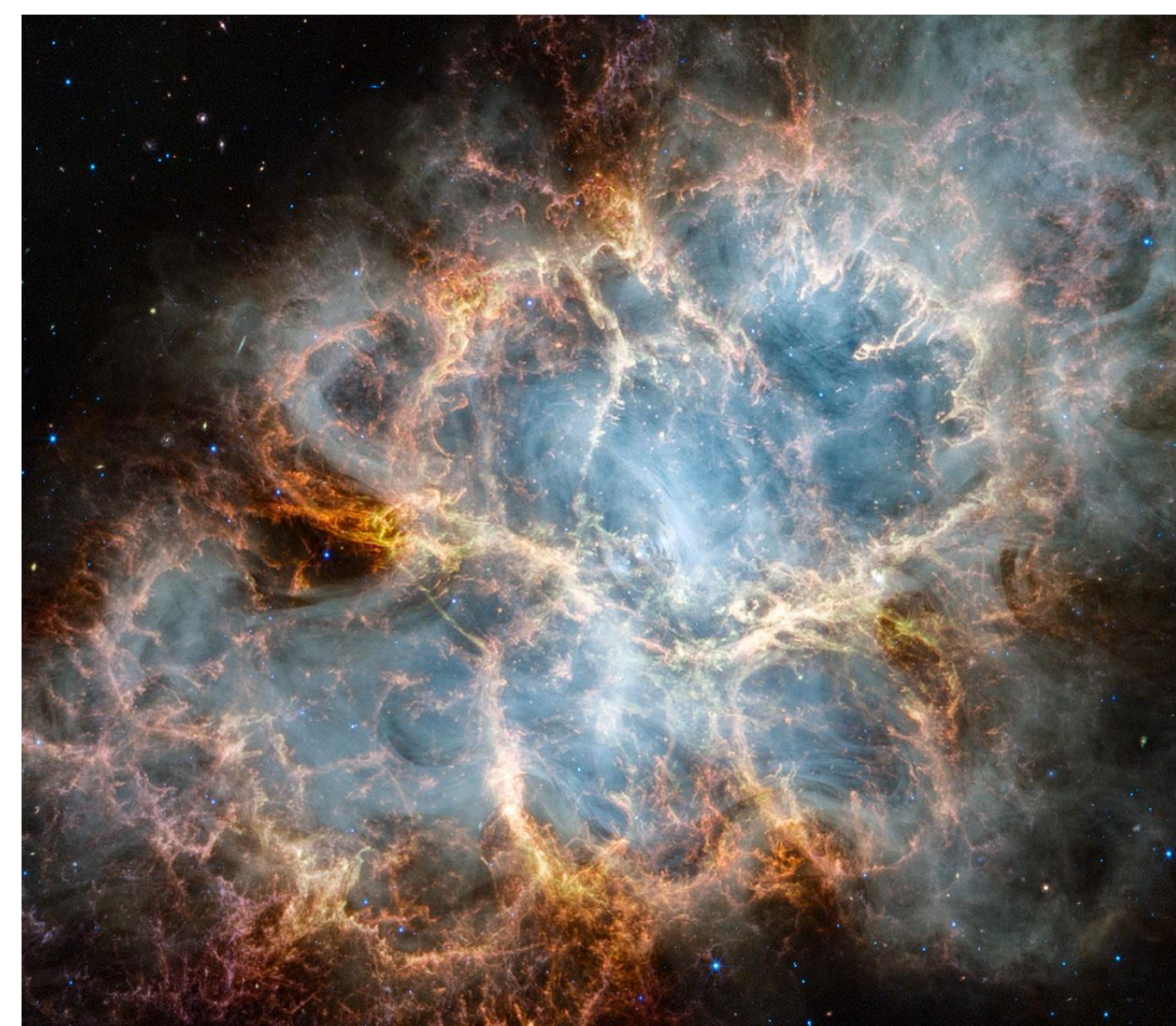
Gammapy method that displays available events and IRFs for each observation



Gamma-ray Event Reconstruction

$\gamma\pi$ A Python package for
gamma-ray astronomy

Gammapy is an open-source Python package for gamma-ray astronomy built on Numpy, Scipy and Astropy. Gamma-ray observatories have traditionally relied on proprietary data and instrument-specific software. Next generation open facilities such as Cherenkov Telescope Array Observatory (CTAO) will make data and analysis tools more broadly accessible.⁴ Gammapy is therefore expanding use cases in the analysis of instruments including Fermi-LAT, H.E.S.S., MAGIC, VERITAS, and HAWC.⁵



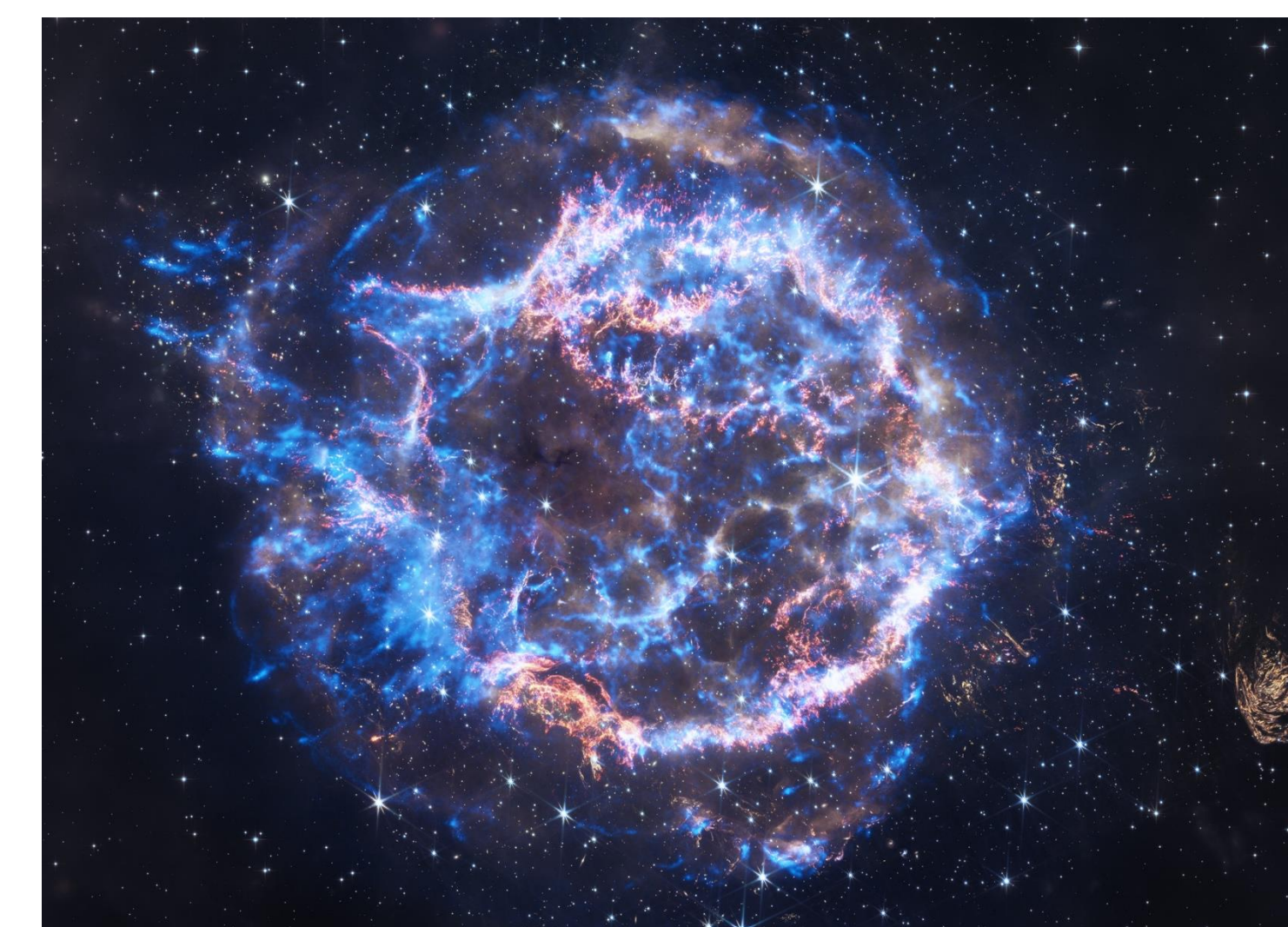
Crab Nebula by James Webb Space Telescope in the infrared red wavelengths

Crab Nebula

- A supernova remnant (SNR) and pulsar wind nebula in the constellation of Taurus.⁶
- Brightest source in the very-high-energy (VHE) gamma-ray sky.⁷
- Widely used as standard candle across X-ray, high-energy, VHE gamma-ray astronomy.⁸
- Often used as calibration source for instruments and analysis pipeline.

Cassiopeia A

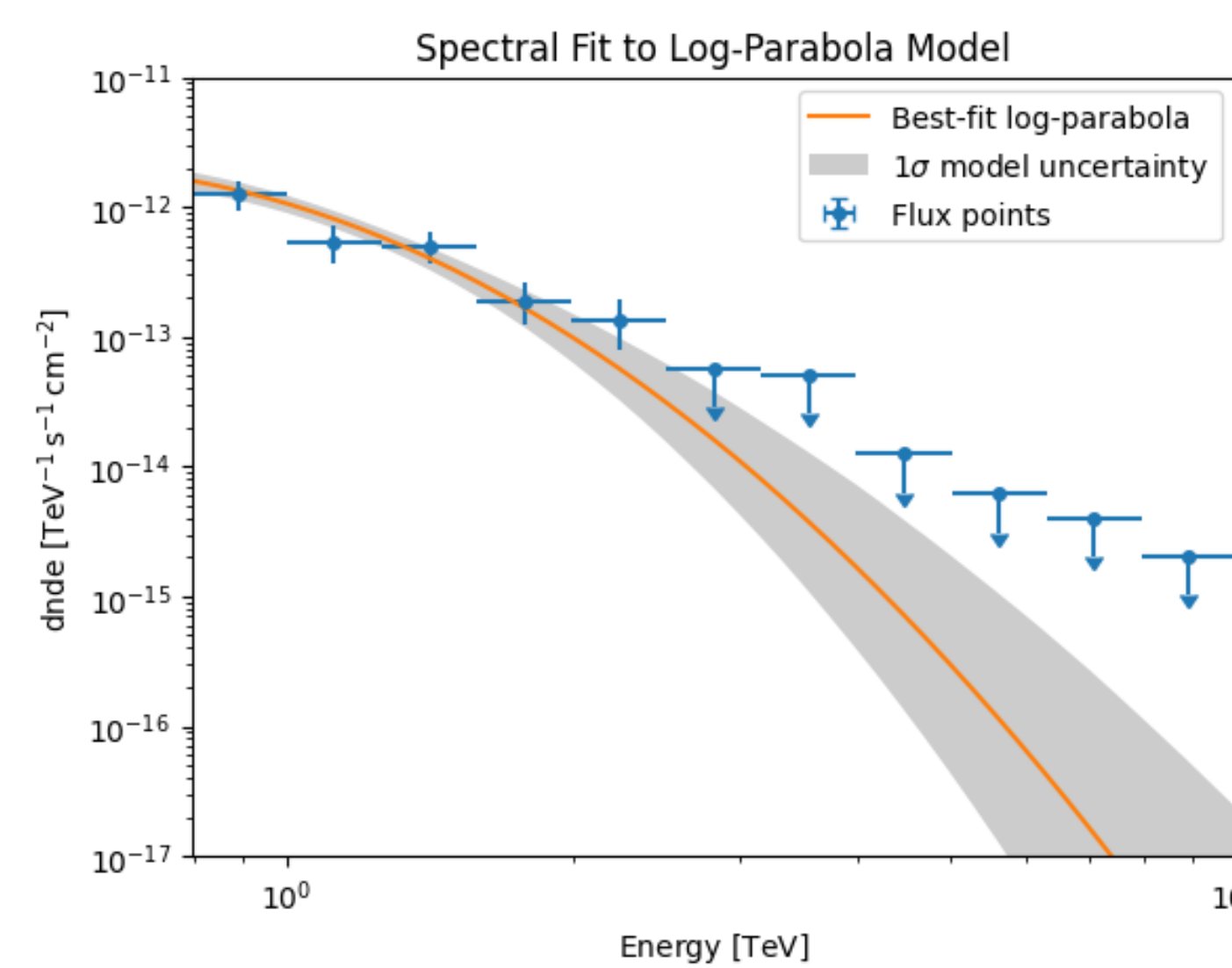
- One of the youngest SNR in the Milky Way.⁹
- The brightest extrasolar radio source, powered by synchrotron process.⁹
- Confirmed to be asymmetric by light echoes study.¹⁰
- Well studied by the VERITAS collaboration.⁹



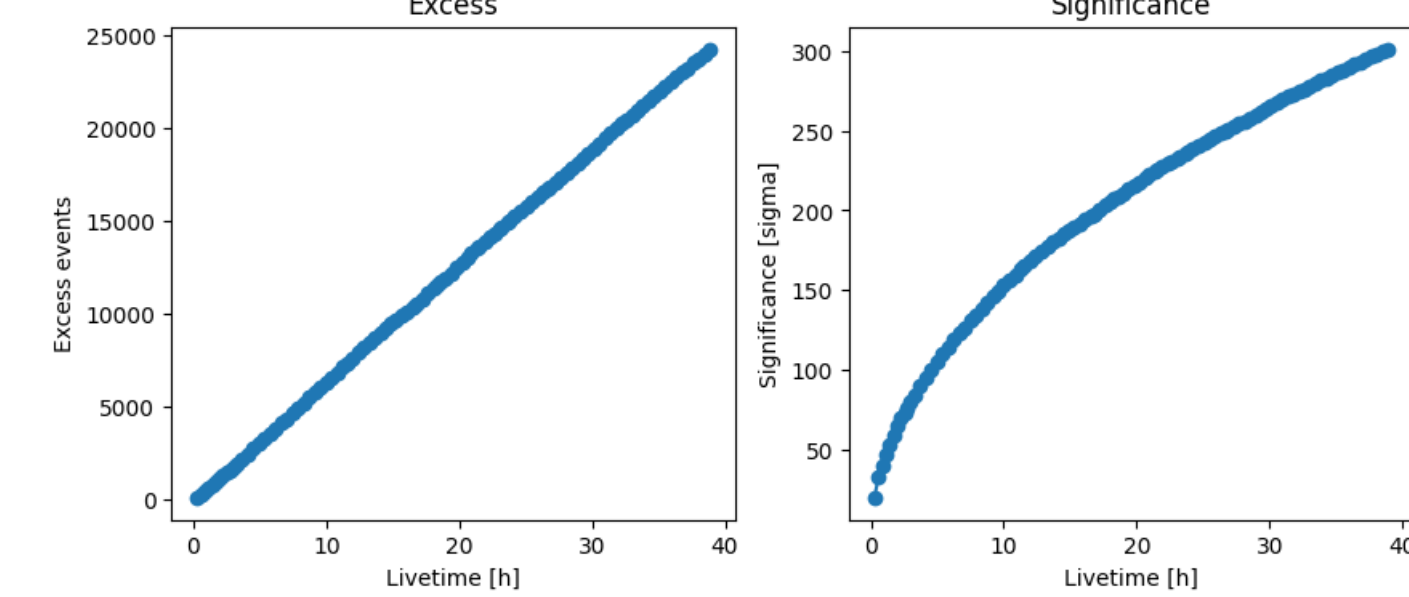
Cassiopeia A by Chandra X-ray Observatory and James Webb Space Telescope

Results

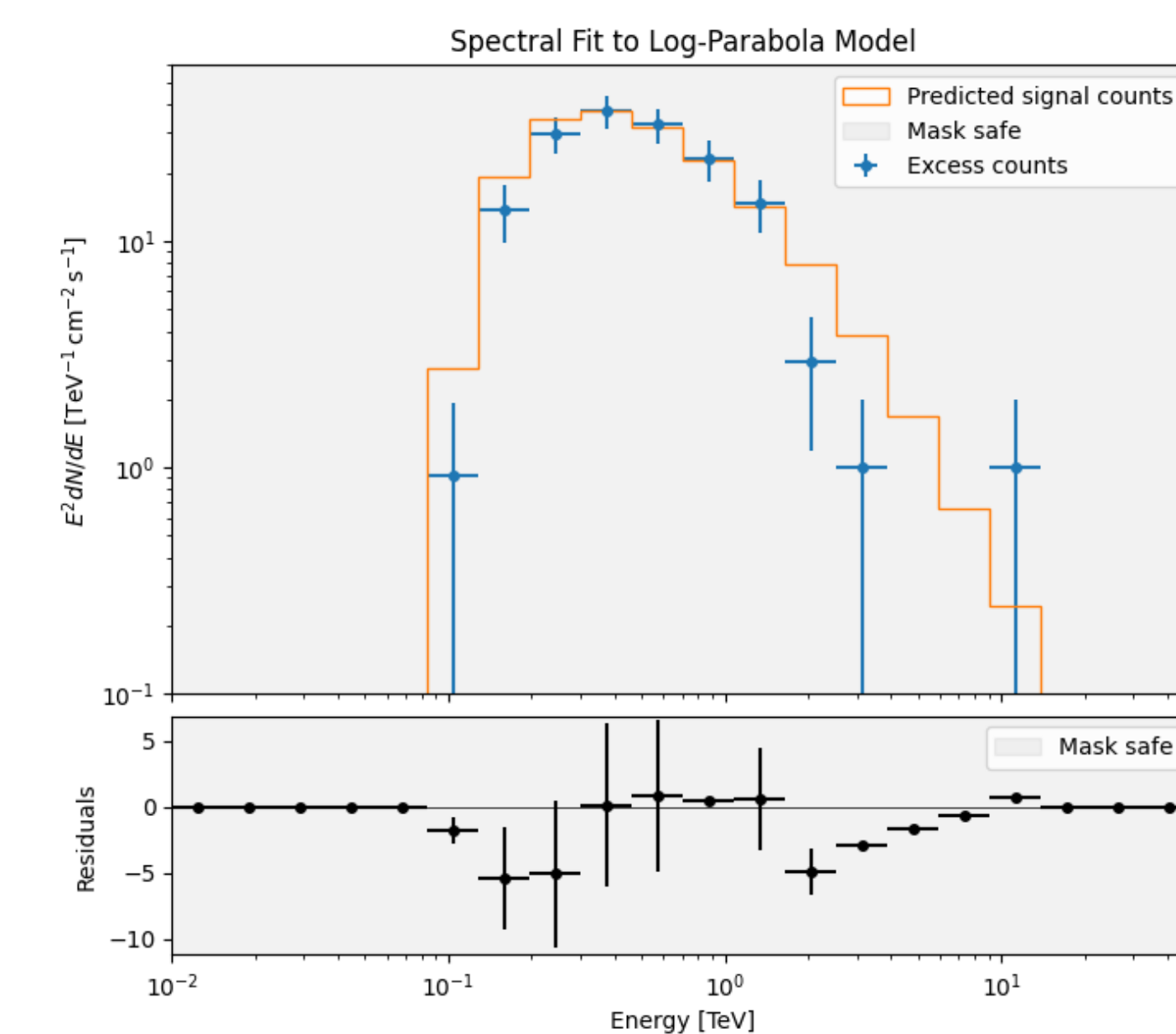
Both analyses closely follow the approaches reported by VERITAS. However, there are several differences in the datasets and methods used. In particular, the Crab Nebula's analysis uses more recent observational data, and the Cassiopeia A analysis uses the Reflected Background Method rather than the Ring Background Method for background modelling. We are interested in reconstructing the spectrum by fitting a log-parabolic function to the measurements.



$$\frac{dN}{dE dA dt} = f_0 \left(\frac{E}{E_0} \right)^{-\alpha - \beta \log(E/E_0)}$$



Above: Excess and Significance Plot
Left: Flux Points Estimator for Cassiopeia A
Right: Global scaled fit of one observation for Crab



Crab Nebula	Normalization f_0 (TeV)	Photon Index α	Curvature Parameter β
VERITAS reported values	$3.7\text{e-}11 \pm 0.03\text{e-}11$	2.4 ± 0.006	0.16 ± 0.01
Reproduced Value	$3.2\text{e-}11 \pm 3.2\text{e-}13$	$2.4 \pm 1.08\text{e-}02$	$0.16 \pm 8.01\text{e-}03$

Cassiopeia A	Normalization f_0 (TeV)	Photon Index α	Curvature Parameter β
VERITAS reported values	$1.45\text{e-}12 \pm 0.11\text{e-}12$	$2.75 \pm 0.10_{\text{stat}} \pm 0.20_{\text{sys}}$	-
Reproduced Value	$1.06\text{e-}12 \pm 1.6\text{e-}13$	2.2 ± 0.4	1.8 ± 0.68

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